
Plot Twists Over Time: How Movie Stories Have Changed Over 95 Years

Ansel Cheung^{*1} Alessio Villa^{*2} Bartol Markovinović^{*3} Martín López de Ipiña^{*4} Niklas Abraham^{*5}

Abstract

Cinema has been a medium for storytelling through visualization, and the evolution of films could tell us about the evolution of culture and societal values. Many sought to quantify this evolution using older natural language methods. We contribute to this ongoing film evolution analysis by scraping and combining multiple sources of data (Wikipedia, IMDB, TMDb) from 1930 to 2024 and embedding the plots with a language model. Using distance distributions, new novelty score and Kolmogorov-Smirnov tests, we show that movie text embeddings are temporally stable with the exception of specific subgenres which demonstrate how movie production changes with respect to cultural and historical contexts.

1. Introduction

The movie industry has evolved considerably over the past century, both because of cultural changes and technological advancements. This raises questions about how historical events influence movie stories, how genres have changed over time, and whether movies are becoming less novel. To address these questions, we collect a large dataset of movie characteristics, including plot summaries from Wikipedia, and perform a time-based analysis of the movie plots.

Previous approaches explore originality and temporal trends in movie plots by employing keyword search or topic modelling (Dubourg et al., 2023). While these older methods might be insightful yet challenging, recent computational work (Xu et al., 2020) has revealed hidden cultural patterns in large narrative corpora. One such analysis conducted on the musical domain uses high dimensional embeddings to

observe changes in structural properties over time (Di Marco et al., 2025). We build upon these foundations by leveraging advances in large language models (LLMs) to embed movie plot summaries (Sreenivasan, 2013) into a unified semantic space. Using distances between these embeddings, we investigate relationships between movie genres across time, and develop a novelty score to quantify the originality of movies. Furthermore, we are able to detect subtle shift in the movie narratives, e.g. a shift away from Spy (James Bond-like) movies after the end of the Cold War.

2. Data and Methods

2.1. Data Collection

The initial dataset was constructed by querying Wikidata, an open knowledge graph structured as a network of items, connected together by properties. We used its SPARQL API to extract items classified as films with release dates ranging from 1930 to 2024. We first retrieved QIDs of items describing feature films, then processed them in small batches to gather the rest of relevant features including title, release date, duration, genres, directors, actors, English Wikipedia link, and more significantly, links to external movie databases TMDb and IMDb. During collection of QIDs, items classified as short film or television series episode were excluded, and final filtering of non-feature films was performed during data cleaning, based on the Wikidata class property.

In the second stage, we enriched the dataset with TMDb statistics, retrieving vote counts, vote averages, and popularity for each film. These served as proxies for audience engagement and informed later filtering and weighting.

The third stage focused on obtaining full-text plot summaries. Leveraging Wikipedia sitelinks from Wikidata, we accessed each film’s Wikipedia page to extract the plot section. Wikipedia’s editorial standards ensure relatively uniform and neutral plot descriptions, facilitating standardized comparative semantic analysis. This step used the Wikipedia API for article retrieval, section extraction, and text normalization, transforming metadata into the dense textual data required for downstream embedding.

The last enrichment stage addressed limitations in TMDb voting statistics. Many movies in the corpus lacked TMDb

^{*}Equal contribution ¹Matrikelnummer 7274374, MSc Machine Learning ²Matrikelnummer 7306912, MSc Computer Science ³Matrikelnummer 7324790, MSc Machine Learning ⁴Matrikelnummer 7293076, MSc Machine Learning ⁵Matrikelnummer 7307188, MSc Machine Learning. Correspondence to: NA <niklas-sebastian.abraham@student.uni-tuebingen.de>.

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ratings or vote counts. To address this issue, we enriched the dataset with IMDb vote averages and vote counts, obtained from IMDb’s non-commercial data files and merged using the IMDb title identifier. The inclusion of IMDb data ensures broader coverage and higher vote volumes, resulting in a more stable measure of audience reception.

All data sources are open and appropriately licensed. Wikidata (Wikimedia Foundation, 2024a) is released under CC0 1.0 Universal (public domain). Wikipedia (Wikimedia Foundation, 2024b) is under CC BY SA 4.0, and TMDb (TMDb, 2024) under CC BY NC 4.0, allowing non-commercial research with attribution.

After data processing, our final dataset consisted of 141,119 movies with 81% average feature coverage.

2.2. Data cleaning

After collecting the raw movie data from Wikidata, TMDb and Wikipedia, we first ensured that our dataset is free of duplicates with respect to Wikidata QIDs and Wikipedia links, and then applied filtering steps described in Figure 1.

A critical step was the removal of movies with low-entropy plots. The raw dataset contained a significant number of incomplete or overly brief plots. To identify and remove such movies we employed a filtering method inspired by (Wenzek et al., 2020), who used perplexity of a LLM to filter out low quality documents. While (Wenzek et al., 2020) discarded documents with low LLM perplexity score, we tokenized the movie plots with the BGE-M3 tokenizer and filtered out the samples with low Shannon entropy of the token distribution. To determine the optimal entropy threshold, we sampled 150 movies from the borderline entropy region of [4.0, 5.5] and manually annotated them as either *good* or *bad* quality. The threshold of 4.8398 was chosen to maximize the $F\beta$ score with $\beta = 0.5$ prioritizing precision over recall.

Embedding of the movie plot summaries. For embedding, we selected the BGE-M3 model (Chen et al., 2024) based on its strong showing (28th place) on the Massive Text Embedding Benchmark (MTEB) leaderboard¹. BGE-M3 is compact (0.5B parameters), supports 8,192-token context length, and can embed entire movie plots in a single vector.

We used a single static model for all text to ensure unified representation in a common latent space. Since Wikipedia plots are modern English (maintained since 2001), linguistic drift is negligible. As embeddings are done token-level, we evaluated chunking and pooling methods to create document embeddings (Devlin et al., 2019; Gong et al., 2019; Raffel et al., 2020) and ended up using the CLS Token (Group,

¹<https://huggingface.co/spaces/mteb/leaderboard>

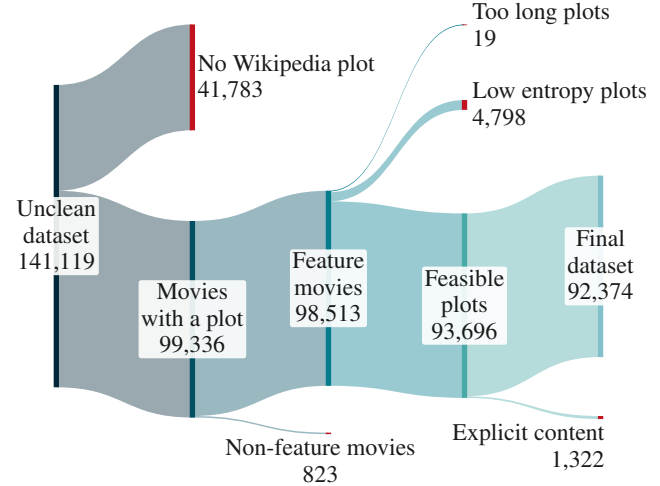


Figure 1. Data cleaning pipeline: Filtering steps with remaining movie counts after each stage. Long plots are defined by Wikipedia as those exceeding 14,000 characters.

2025b).

Internal data validation. To check that the embeddings meaningfully represent plot similarity, we measured cosine distances within and outside major movie franchises (Harry Potter, Star Wars, James Bond). Movies from the same franchise consistently clustered closer together (e.g., Harry Potter: 0.21 inner-group distance vs. 0.57 to random movies), while the global average distance between any two movies was 0.52. This confirms that the embeddings capture true semantic similarity.

Genre Taxonomy. The raw dataset included 975 unique genre labels, many of which were redundant or highly similar. To simplify and standardize the taxonomy, we first removed genres that appeared only once, reducing the set to 463. We then embedded the Wikipedia descriptions (available for 359 genres) using the BGE-M3 model, and clustered these vectors with $k = 20$ using k-means. Each cluster was manually labeled based on thematic similarity, resulting in 20 coherent genre categories used in further analyses.

3. Methodology

With data collected and cleaned, we embed movie plot summaries into a semantic space and analyze the embeddings using three complementary approaches: distance-based drift analysis, novelty scoring, and Kolmogorov-Smirnov tests.

Distance analysis. Once movie plots are embedded into a unified semantic space, quantitative analysis of their geometric relationships becomes possible through distance metrics. The cosine distance between embeddings provides a natural measure of semantic dissimilarity, enabling the construction

of cumulative distribution functions over pairwise distances within defined subsets of the corpus.

As movie genres provide a meaningful taxonomy with potential temporal evolution patterns, we examine semantic drift across different time periods of an arbitrary number of years. To this end, embeddings are first grouped by genres into decades. For each group, two alternative representative embeddings are computed: the **centroid** (arithmetic mean) and the **medoid** (cosine distance minimizer embedding). We employ the following metrics to analyse the drift dynamics across the groups: (i) genre drift and acceleration, and (ii) inter-genre distance.

Genre drift measures the cosine distance between representative embeddings of consecutive periods, capturing how much a genre’s semantic center evolves over two time periods. Acceleration quantifies the change in drift between consecutive periods.

Inter-genre distance determines cosine distance between representatives of each pair of genres for each time period, enabling pairwise comparison between specific genres.

Since our dataset has significantly more movies in recent years, the medoid and centroid embedding estimators are heavily affected by sampling variance. To mitigate this, we downsample to ensure equal sampling error across all genre-year groups and apply bootstrapping to estimate confidence intervals.

Novelty analysis. To investigate the claim that movies are becoming less novel over time, we propose a novelty metric defined as the minimal cosine distance between a specific movie’s plot embedding and the embeddings of all movies in the dataset released prior to it. This can be formally written as:

$$\text{Novelty}(m_i) = \min_{j:\text{year}_j < \text{year}_i} \left(1 - \frac{\text{emb}_i \cdot \text{emb}_j}{\|\text{emb}_i\| \|\text{emb}_j\|} \right) \quad (1)$$

where emb_i denotes the embedding vector of i -th movie’s plot. Intuitively, a higher novelty score indicates that the movie’s plot is more dissimilar from prior movies, while a lower score implies existence of a very similar movie released earlier. To compute these scores, we leverage the Faiss library (Douze et al., 2024) to efficiently find the preceding nearest neighbors of movie plot embeddings based on the cosine distance.

Kolmogorov-Smirnov test. To compare distance distributions across movie subsets, we use the Kolmogorov-Smirnov test (Massey, 1951). We construct ϵ -balls ($\epsilon \in [0.24, 0.30]$) around anchor movies representing specific themes, collecting all movies within the distance threshold to define local semantic neighborhoods. We compare distance distributions and temporal CDFs (cumulative distribution functions of release years) within these neighborhoods to a control group

(global mean embedding). Divergence indicates temporal shifts: if semantic structure remains stable, distributions should be similar across periods.

4. Results

General Spatial Analysis. We begin by examining the global structure of the embedding space through pairwise cosine distances between movies. These distances are approximately normally distributed, with a mean of $\mu = 0.5195$ and a standard deviation of $\sigma = 0.0624$, capturing the typical dissimilarity between movie plots and serving as a baseline for further analysis.

To evaluate whether genre labels map to distinct regions, we compared intra-genre (mean: 0.5042) and inter-genre (mean: 0.5268) distances across 19 genres. The resulting separation gap of 0.0226, ratio of 1.0448, and a silhouette score of -0.0334 indicate considerable overlap, with genres only weakly separated in the embedding space. This suggests genre boundaries are porous, consistent with the hybrid and overlapping nature of film categories.

Genre drift analysis. After normalization, the cosine distance of each time group representative embedding with respect to the previous one exhibited only random fluctuation with no clear trend; inter-genre analysis yielded the same result. These outcomes suggest that genres may be too broad as analytical categories, with any underlying patterns likely obscured by noise.

Spread analysis. We analyzed the spread of movies per year using three metrics: mean L2 norm, Frobenius norm, and spectral norm of each movie embedding centered by its respective yearly centroid (i.e. yearly centroid was subtracted from each movie embedding before computing the norms) (Yamagiwa & Shimodaira, 2024). As with the drift analysis, no interpretable patterns emerged: mean L2 norm and Frobenius norm stayed relatively constant at 0.7 and 12.4 respectively, whereas spectral norm had a slight increase from 2.1 to 2.8. This indicates that the overall spread of movies remains relatively constant over the years with outliers becoming more polarizing. The nature of these polarizing axes proved difficult to characterize, as they changed yearly and were a combination of multiple dimensions.

Novelty score. To judge whether movies are becoming less novel over time, we computed novelty scores as defined in 1, and plotted them against release years in Figure 2.

Fitting a linear regression to the average yearly novelty scores of all movies (slope of $3.3 \cdot 10^{-4}$ and R^2 value of 0.56) indicates that the yearly averages have stayed constant over time. Additionally, we compared these averages to the yearly averages of Best Original Screenplay Oscar nominees by computing the Mean Absolute Error (MAE) between the

two time series. The resulting MAE of 0.0098 ± 0.008 suggests that the scores of nominees don't significantly differ from the yearly novelty averages of all movies.

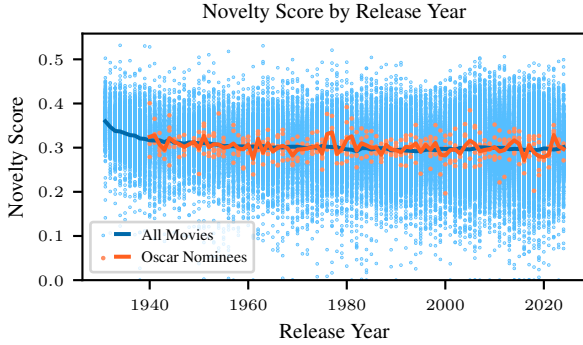


Figure 2. Novelty scores of all movies (blue) and Best Original Screenplay Oscar nominees (orange) over time. Each dot represents a movie and the lines represent yearly average novelty scores.

Kolmogorov-Smirnov test. We demonstrate our approach by using James Bond films as anchor movies and comparing their distance distributions to those of all other movies. The cumulative distribution of cosine distances from the Bond anchors rises steeply only for a small subset of closely related films, while most movies are more distant. To analyze temporal trends, we constructed cumulative distribution functions of release years for movies within the ϵ -ball around the anchors and compared them to a control group. Figure 3 shows a difference in temporal distributions. The left panel reveals a divergence beginning around the 1960s, indicating that the spy movie subgenre represented by the Bond anchor displays a distinctive pattern of temporal emergence compared to the broader corpus. The right panel displays normalized histograms of movie counts per year for both groups, confirming that the temporal distribution of spy-themed films differs from that of all movies overall. This temporal divergence suggests that the spy film subgenre experienced a period of increased production and thematic consolidation that is not seen in general cinematic trends of the same period. We conducted the same analysis for other subgenres, such as movies about American soldiers in the Middle East, time travel, and the Wild West. In each case, a clear pattern shift is observed; for example, the American soldiers in the Middle East subgenre began to emerge in the 1990s and grew after 2001, while Wild West movies have steadily declined since the 1980s.

5. Discussion & Conclusion

We have utilized modern embedding methods and multiple statistical tools to analyze the evolution of movie plot embeddings over nearly a century of cinema. Our findings indicate that while the overall semantic structure of movie plots remains relatively stable, there were declines and growth

K-S Test: Temporal Distributions ($\epsilon=0.29$)
K-S Statistic: 0.109942

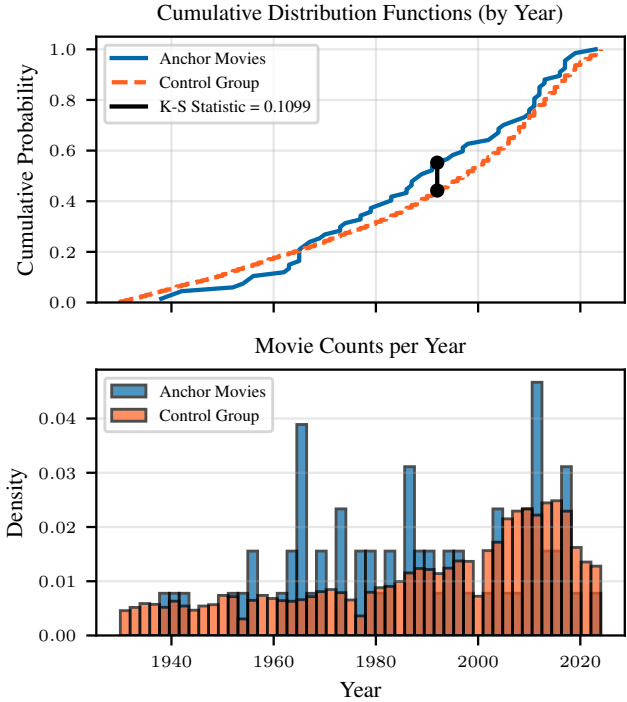


Figure 3. KS test of temporal distributions for James Bond epsilon ball ($\epsilon = 0.29$) versus control group, showing temporal divergence. Left: Cumulative distribution functions. Right: Normalized yearly counts.

of subgenres which were statistically sound and could be explained by real world occurrences, e.g. the decline in number of spy movies after the Cold War.

We must acknowledge the limitations that arise from temporal analysis using text embeddings. Wikipedia plot summaries were already temporally normalized and may not fully capture the essence of the actual movie, potentially introducing bias of the article authors. Additionally, using a single embedding model, while ensuring a unified semantic space, may overlook temporal linguistic shifts. More importantly, the evolution of cinema is not only reflected in plot summaries but also in cinematography, direction, music, feeling, acting performance, basically other non-textual elements. Future work could explore multimodal embeddings that combines visual and auditory features alongside textual data.

All code and data used in this project are openly available in our GitHub repository (Group, 2025b) and as a dedicated dataset on Hugging Face (Group, 2025a).

Contribution Statement

Contribution Statement:

- **Ansel Cheung:** Performed genre classification analysis, classification of movie plots into genres, and conducted spread and PCA analysis of the movie plots.
- **Alessio Villa:** Developed and maintained the IMDb and TMDb API pipelines, and contributed to the related work research and methods background sections.
- **Bartol Markovinović:** Defined the data pipeline cutoff and carried out resulting data cleaning, managed the integration of Wikidata, and conducted novelty score analysis.
- **Martín López de Ipiña:** Carried out genre drift statistical analysis on the general embedding space, performed general spatial analysis of embeddings, and analyzed the cosine distance distributions.
- **Niklas Abraham:** Performed embedding model selection and evaluation, analyzed chunking methods, and performed KS test and distance distribution analysis.

Overall, all authors contributed equally to the project. This is reflected in the various analysis sections throughout the report, where each member’s work formed an integral and balanced part of the final study.

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